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Homework 3

$$1. (a): f(\theta) = (1 - [1, 0.1, 0.01]\theta)^2$$

$$f'(\theta) = 2(1 - [1, 0.1, 0.01]\theta) \cdot \frac{\partial (1 - [1, 0.1, 0.01]\theta)}{\partial \theta}$$
$$= 2(1 - [1, 0.1, 0.01]\theta) \cdot \begin{bmatrix} 1 \\ 0.1 \\ 0.01 \end{bmatrix}$$

Let $f'(\theta) = 0$, that's when $1 - [1, 0.1, 0.01]\theta = 0$

$$\therefore \textcircled{1} \theta_1 + \frac{1}{10}\theta_2 + \frac{1}{100}\theta_3 = 1$$

$\therefore 2(1 - [1, 0.1, 0.01]\theta)$ is a constant

$\therefore$ The update direction of θ is $\begin{bmatrix} 1 \\ 0.1 \\ 0.01 \end{bmatrix}$

$$\therefore \theta_0 = 0$$

$$\therefore \textcircled{2} \theta_1 = 10\theta_2 = 100\theta_3$$

Substitute $\textcircled{2}$ into $\textcircled{1}$: $\theta_1 + \frac{1}{10}\theta_1 + \frac{1}{10000}\theta_1 = 1$

$$\therefore \theta_1 = \frac{10000}{10101}, \theta_2 = \frac{100}{10101}, \theta_3 = \frac{1}{10101}$$

$$\text{To conclude: } \theta = \begin{bmatrix} \frac{10000}{10101} \\ \frac{100}{10101} \\ \frac{1}{10101} \end{bmatrix}$$

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1. (b): I have proved in (a) that $g_t = m \cdot \begin{bmatrix} 1 \\ 0.1 \\ 0.01 \end{bmatrix}$ (m is a constant)

$$M_t g_t = \begin{bmatrix} \frac{1}{m} & 0 & 0 \\ 0 & \frac{10}{m} & 0 \\ 0 & 0 & \frac{100}{m} \end{bmatrix} \cdot \begin{bmatrix} m & 0 & 0 \\ 0 & 0.1m & 0 \\ 0 & 0 & 0.01m \end{bmatrix}$$

$$= \text{sign}(m) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \Delta\theta_1 = \Delta\theta_2 = \Delta\theta_3$$

$$\because \theta_0 = 0$$

$$\therefore \textcircled{1} \theta_1 = \theta_2 = \theta_3$$

Substitute ① into $[1 \ 0.1 \ 0.01]\theta = 1$

We can get $1.11\theta_1 = 1$

$$\therefore \theta_1 = \theta_2 = \theta_3 = \frac{100}{111}$$

Therefore, θ will converge to $\frac{100}{111} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

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1. (c):

$$S_1 = X_{11} = 1, \quad S_2 = X_{12} = 0.1, \quad S_3 = X_{13} = 0.01$$

$$\text{Let } S = \begin{bmatrix} S_1 & 0 & 0 \\ 0 & S_2 & 0 \\ 0 & 0 & S_3 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 0.1 \\ 0.01 \end{bmatrix}$$

$$\therefore y = x^T \theta, \quad \hat{x} = S^T x$$

$$\therefore y = (S \hat{x})^T \theta = \hat{x}^T S^T \theta = \hat{x}^T \hat{\theta}$$

$$\therefore \hat{\theta} = S^T \theta$$

$$\therefore \hat{x} = S^T x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 100 \end{bmatrix} \begin{bmatrix} 1 \\ 0.1 \\ 0.01 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore f(\hat{\theta}) = (1 - \hat{x}^T \hat{\theta})^2$$

$$f'(\hat{\theta}) = 2(1 - \hat{x}^T \hat{\theta}) \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = m \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (m \text{ is a constant})$$

$$\therefore \hat{\theta}_0 = 0$$

$$\therefore \hat{\theta}_1 = \hat{\theta}_2 = \hat{\theta}_3$$

$$\therefore 1 = [1 \ 1 \ 1] \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \\ \hat{\theta}_3 \end{bmatrix}$$

$$\therefore \hat{\theta} \text{ converges to } \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$\therefore \theta = (S^T)^{-1} \hat{\theta}$$

$$\therefore \theta \text{ converges to } \frac{1}{3} \begin{bmatrix} 1 \\ 10 \\ 100 \end{bmatrix}$$

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$$\begin{aligned} 2. \|w_t\|_2 &= \|w_{t-1} - \eta(F^T(Fw_{t-1} - y))\|_2 \\ &= \|(I - \eta F^T F)w_{t-1} + \eta F^T y\|_2 \\ &\leq \|(I - \eta F^T F)w_{t-1}\|_2 + \eta \|F^T y\|_2 \\ &\leq \sigma_{\max}(I - \eta F^T F) \|w_{t-1}\|_2 + \eta \sigma_{\max}(F^T) \|y\|_2 \end{aligned}$$

$\because \eta$ is small enough

$$\therefore \sigma_{\max}(I - \eta F^T F) = 1 - \eta \sigma_{\min}^2 < 1$$

$$\because \sigma_{\max}(F^T) = \alpha$$

$$\therefore \sigma_{\max}(I - \eta F^T F) \|w_{t-1}\|_2 + \eta \sigma_{\max}(F^T) \|y\|_2 \leq \|w_{t-1}\|_2 + \eta \alpha \|y\|_2$$

$$\text{Therefore, } \|w_t\|_2 \leq \|w_{t-1}\|_2 + \eta \alpha \|y\|_2$$

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4. $n(n-1)$, P , $\frac{P}{n}$